

IP address

network configuration on modern Linux systems. To set a static IP:

```
nmcli connection modify eth0 ipv4.addresses
192.168.1.100/24
nmcli connection modify eth0 ipv4.gateway
192.168.1.1
nmcli connection modify eth0 ipv4.dns
8.8.8.8
nmcli connection modify eth0 ipv4.
method manual
nmcli connection up eth0
```

Here, the `ipv4.addresses` field sets the static IP, while `ipv4.gateway` defines the default route. The `ipv4.dns` field specifies the DNS server, ensuring domain name resolution.

Configuring DHCP with Netplan

For Ubuntu and Debian-based systems using Netplan, DHCP configuration is straightforward.

Edit the configuration file (e.g., `/etc/netplan/01-netcfg.yaml`):

```
network:
  ethernets:
    eth0:
      dhcp4: true
  version: 2
```

Apply changes with:

```
sudo netplan apply
```

This configuration tells the system to obtain an IP address dynamically from a DHCP server.

Network Interfaces and Routing (`ip`, `route`)

Understanding network interfaces and routing is essential for directing traffic efficiently. Linux provides the `ip` command to manage network interfaces, IP addresses, and routes.

List interfaces:

```
ip link show
```

Bring an interface up or down:

```
ip link set eth0 up
```

```
ip link set eth0 down
```

Display routing table:

```
ip route show
```

Add a static route:

```
ip route add 192.168.2.0/24 via 192.168.1.1
dev eth0
```

Routes are crucial for directing network traffic efficiently. For instance, a static route might be required when dealing with multiple subnets.

DNS Configuration and Troubleshooting

DNS (Domain Name System) resolves human-readable domain names into IP addresses. Without proper DNS configuration, accessing websites and remote servers by domain name would be impossible.

To manually configure DNS, edit `/etc/resolv.conf`:

```
nameserver 8.8.8.8
nameserver 1.1.1.1
```

Test DNS resolution:

```
nslookup google.com
dig google.com
```

When troubleshooting DNS, use tools like `nslookup` and `dig` to verify name resolution and check the status of DNS servers.

Remote Access and Security

SSH: Key-Based Authentication and Config Files

Secure Shell (SSH) enables remote access to Linux systems. Key-based authentication enhances security by eliminating the need for password-based logins.

Setting up SSH Key Authentication

```
ssh-keygen -t rsa -b 4096
ssh-copy-id user@remote-host
```

The above commands generate an RSA key pair and copy the public key to the remote host. This ensures secure password-less authentication.

To further secure SSH, disable password authentication in `/etc/ssh/sshd_config`:

```
PasswordAuthentication no
```

Restart SSH to apply changes:

```
sudo systemctl restart ssh
```

Firewalls: `iptables`, `nftables`, and `firewalld`